IBM AI Racing League

Windows Setup Guide

Winter 2026 Season — native installation, no containers required

This guide walks you through setting up the full IBM AI Racing League environment directly on Windows. Follow the steps in order.

Step	What you will do
1. Prerequisites	Hardware requirements, disk space, Python 3, Git, C++ Build Tools
2. TORCS + SCR	Download and unzip — no installer, just extract and run
3. Python packages	Create a virtual environment and install necessary libraries
4. gym_torcs	Set up the AI starter kit provided by the organizers
5. IBM Bob	Install the IBM fork of the VS Code editor with the IBM AI assistant and set up the Python extension

Downloads

Windows Installation Files (just a zip file; contains the torcs simulator and the gym_torcs starter kit)

Prefer a simpler setup?
If you would rather skip manual installation, use the pre-made container image with everything baked in (TORCS, patches and IBM Bob) — see the Docker Setup Guide.

1. Prerequisites

1.1 Hardware requirements

Requirement	Details
OS	Windows 10 or Windows 11 (64-bit)
CPU	Any modern 64-bit processor. TORCS is lightweight — designed for 800 MHz CPUs, so any current machine is more than sufficient.
RAM	4 GB minimum. 8 GB recommended when running TORCS and the Python agent simultaneously. 16 GB if you plan to train neural networks.
Graphics card	Any GPU with OpenGL 1.3 support — this covers virtually every card made in the last 20 years including integrated Intel/AMD graphics. A dedicated GPU is not required for TORCS. An NVIDIA GPU with CUDA is only needed if you want faster neural network training.
Disk space	12 GB free (see breakdown below)
Display	Any monitor — TORCS runs in a standard window

1.2 Disk space breakdown

Component	Space needed
TORCS + gym_torcs (extracted zip)	~1 GB
Python 3	~100 MB
Visual C++ Build Tools	~4 GB
IBM Bob (VS Code-based editor)	~400 MB
Python virtual env + packages (PyTorch, TensorFlow, etc.)	~3-4 GB
Working space during installation	~2 GB
Total recommended free space	12 GB

No dedicated GPU needed for this competition

TORCS renders 3D graphics but its requirements are very modest — integrated graphics on any laptop from the last 10 years will run it fine. A dedicated GPU only matters if you want to speed up neural network training, which is optional.

1.3 Software prerequisites

Install these three tools before starting. All use standard Windows installers.

1.4 Python 3

1. Go to https://www.python.org/downloads/ and download the latest Python 3 installer.
2. Run the installer. On the first screen tick 'Add Python to PATH' before clicking Install Now — this is critical.
3. Open PowerShell (press Windows key, type PowerShell, press Enter) and verify:

```
python --version
```

Expected output: Python 3.12.x or similar.

1.5 Git for Windows

1. Go to https://git-scm.com/download/win and download the installer.
2. Run with all default settings.
3. Verify in PowerShell:

```
git --version
```

1.6 Visual C++ Build Tools

Required for compiling some Python packages. Download from:

```
https://visualstudio.microsoft.com/visual-cpp-build-tools/
```

1. Run the installer, select the 'Desktop development with C++' workload, and click Install.

2. TORCS + SCR Patch

TORCS on Windows runs directly from a folder — no formal installation needed. The SCR patch (the UDP socket interface your AI connects to) and the competition car designs are all included in the zip file provided by the organisers.

2.1 Download

1. Download the Windows Installation Files (a single zip file) from the link below. It contains both the TORCS simulator and the gym_torcs starter kit:

```
https://ibm.box.com/s/adts01qbpa6kj9g7oao1bb87pg9lfns4
```

2.2 Extract

1. Right-click the downloaded zip and select Extract All.
2. Extract to a simple path with no spaces. In the destination box type exactly:

```
C:\RaceYourCode\
```

3. The zip contains two folders. After extraction you should have:

```
C:\RaceYourCode\torcs\wtorcs.exe              <- simulator
C:\RaceYourCode\torcs\drivers\scr_server\0\   <- car designs
C:\RaceYourCode\gym_torcs\                    <- starter kit
```

2.3 Test TORCS

1. Open C:\RaceYourCode\torcs\ in File Explorer and double-click wtorcs.exe.
2. The TORCS main menu should appear. If Windows shows a security warning click 'More info' then 'Run anyway'.
3. Close TORCS for now — you will use it again in section 4.

Add TORCS to PATH (optional)
To be able to type 'wtorcs' from any PowerShell window, open PowerShell as Administrator and run: [System.Environment]::SetEnvironmentVariable('Path', $env:Path + ';C:\RaceYourCode\torcs', 'User') — then restart PowerShell.

3. Python Packages

A virtual environment keeps competition packages isolated from the rest of your system.

3.1 Create and activate

```
python -m venv C:\torcs-env
C:\torcs-env\Scripts\activate
```

Your prompt will show (torcs-env) at the start confirming it is active. You need to run the activate command each time you open a new PowerShell window.

3.2 Install packages

With the virtual environment activated:

```
pip install numpy scipy scikit-learn torch tensorflow keras gymnasium
matplotlib pandas
```

GPU / CUDA support
If you have an NVIDIA GPU, visit https://pytorch.org/get-started/locally/ and use the custom PyTorch install command for your CUDA version for faster AI training.

4. gym_torcs Starter Kit

The gym_torcs folder is included in the same zip file you extracted in section 2, so there is nothing to copy or move.

It contains the Python wrapper that connects your AI code to TORCS and the starter agent you will modify.

4.1 Check the folder structure

1. Open C:\RaceYourCode\ in File Explorer. You should see the torcs and gym_torcs folders next to each other.
2. Check that the gym_torcs folder contains these files:

```
C:\RaceYourCode\gym_torcs\torcs_jm_par.py
C:\RaceYourCode\gym_torcs\snakeoil3_gym.py
C:\RaceYourCode\gym_torcs\gym_torcs.py
C:\RaceYourCode\gym_torcs\practice.xml
```

4.2 Key files

File	Purpose
torcs_jm_par.py	Main starter file — this is what you edit for the competition
snakeoil3_gym.py	UDP client — connects Python to TORCS on port 3001. Do not edit.
gym_torcs.py	OpenAI Gym-style RL environment wrapper for advanced users
practice.xml	Race configuration file

4.3 Test the full connection

1. Start TORCS (double-click C:\RaceYourCode\torcs\wtorcs.exe).
2. In TORCS: Race → Practice → Configure Race → select a track → set scr_server as driver → Accept.
3. Race → Practice → New Race. TORCS shows a blue waiting screen — this is correct.
4. Open PowerShell, activate the virtual environment, and run:

```
C:\torcs-env\Scripts\activate
cd C:\RaceYourCode\gym_torcs
```

```
python torcs_jm_par.py
```

5. You should see 'Client connected on 3001' and the car starts moving on track.

Congratulations!
If the car is moving and the terminal shows 'Client connected', the full stack is working. You are ready to start modifying torcs_jm_par.py.

5. IBM Bob

IBM Bob is the IBM fork of the VS Code editor with the IBM AI assistant built in. You will use it to edit your car code and to get AI coding help. It replaces the separate editor and local AI assistant used in earlier seasons.

5.1 Register and install

1. Register for IBM Bob using this link: https://ibm.biz/university-bob
2. Follow the instructions you receive after registering to download the IBM Bob installer for Windows.
3. Run the installer with default settings, then start IBM Bob and sign in with your registered account.

5.2 Install the Python extension

1. Press Ctrl+Shift+X to open Extensions.
2. Search for Python and install the Python extension.

5.3 Set the Python interpreter

1. Press Ctrl+Shift+P and type: Python: Select Interpreter
2. Select the torcs-env at C:\torcs-env\Scripts\python.exe

5.4 Open your competition folder

1. In IBM Bob choose File → Open Folder and select C:\RaceYourCode\gym_torcs
2. Click on torcs_jm_par.py in the Explorer panel to open the main competition file.

5.5 Try the AI assistant

1. Open the IBM Bob AI assistant from the side panel.
2. With torcs_jm_par.py open, ask something like: What does STEER_GAIN do in this file?
3. The assistant should respond with an explanation of the parameter. Use it throughout the season to understand, modify and improve your car code.

6. Your First Modification

Now that everything is working, here is how to make your first change to the AI agent and see the effect on track.

6.1 Understanding the parameters

Open torcs_jm_par.py in IBM Bob. Scroll to the section labelled USER CONFIGURABLE PARAMETERS (around line 493). You will see:

```
TARGET_SPEED = 100       # Target speed in km/h
STEER_GAIN = 30          # Steering sensitivity
CENTERING_GAIN = 0.20    # How strongly car stays centred
BRAKE_THRESHOLD = 0.9    # When to brake in corners
GEAR_SPEEDS = [0, 20, 40, 80, 100, 180]  # Gear shift points
ENABLE_TRACTION_CONTROL = True
```

6.2 Make a change

1. Try changing TARGET_SPEED from 100 to 150:

```
TARGET_SPEED = 150
```

2. Save the file with Ctrl + S.
3. Make sure TORCS is running and waiting (repeat steps 1–3 of section 4.3 if needed).
4. In the terminal, run the agent again:

```
python torcs_jm_par.py
```

5. Watch the car on track. It should now drive faster. Notice whether it stays on track or spins out — you may need to also adjust STEER_GAIN or BRAKE_THRESHOLD to compensate.

Tip: iterate quickly
The typical competition workflow is: change a parameter, save, run the agent, watch the car, stop the agent (Ctrl+C), change again. Each run only takes a few seconds to start. Do not be afraid to experiment.

6.3 What each parameter affects

Parameter	What to try
TARGET_SPEED	Increase for faster lap times, but the car may spin out in corners

STEER_GAIN	Increase if the car is drifting sideways, decrease if it oversteers
CENTERING_GAIN	Increase if the car wanders; decrease if it makes too sharp corrections
BRAKE_THRESHOLD	Decrease (e.g. 0.5) to brake earlier before corners
GEAR_SPEEDS	Adjust shift points for more aggressive or conservative gear changes
ENABLE_TRACTION_CONTROL	Set to False to see what happens without it (spoiler: wheelspin)

7. Troubleshooting

Here are solutions to the most common problems students encounter.

Python script says 'Connection refused' or keeps waiting

This means the Python agent cannot find TORCS. Check the following:

- TORCS must be running and showing the blue waiting screen before you run the Python script
- Make sure you started a New Race in Practice mode with scr_server as the driver
- Run both TORCS and the Python script on the same computer, with the virtual environment activated

The car immediately goes off track and stops

This is normal behaviour with the default parameters on some tracks. The example agent is tuned for moderate-difficulty tracks. Try:

- Reducing TARGET_SPEED to 80 or 60
- Increasing CENTERING_GAIN to 0.4
- Choosing a simpler track in TORCS (a straight or gentle oval)

IBM Bob does not find your Python packages

If imports such as numpy or gymnasium are flagged as missing, IBM Bob is not using your virtual environment. Press Ctrl+Shift+P, run Python: Select Interpreter and choose C:\torcs-env\Scripts\python.exe. Then open a new terminal inside IBM Bob and run the script again.

8. Quick Reference Card

Key files

File	What it is
C:\RaceYourCode\gym_torcs\torcs_jm_par.py	Main starter file — edit this for the competition
C:\RaceYourCode\gym_torcs\snakeoil3_gym.py	Core UDP client — do not edit
C:\RaceYourCode\gym_torcs\gym_torcs.py	RL environment wrapper — advanced users
C:\RaceYourCode\gym_torcs\practice.xml	Race configuration

Key paths

Item	Path
TORCS executable	C:\RaceYourCode\torcs\wtorcs.exe
SCR car designs	C:\RaceYourCode\torcs\drivers\scr_server\
Virtual environment	C:\torcs-env\
Competition code	C:\RaceYourCode\gym_torcs\
IBM Bob registration	https://ibm.biz/university-bob

Key commands (PowerShell)

Task	Command
Activate virtual env	C:\torcs-env\Scripts\activate
Run example agent	python torcs_jm_par.py
Open workspace in IBM Bob	File → Open Folder → C:\RaceYourCode\gym_torcs
Stop the agent	Ctrl + C

Typical session

1. Double-click C:\RaceYourCode\torcs\wtorcs.exe to start TORCS

2. In TORCS: Race → Practice → New Race (scr_server must be the configured driver)
3. Open PowerShell and activate your virtual environment
4. cd C:\RaceYourCode\gym_torcs
5. python torcs_jm_par.py
6. Watch the car, stop with Ctrl+C, edit your code, repeat

Good luck and happy racing!